3NF Explanation Doc

Which transitive dependencies did you find?

- *ProjectID → ClientName → ClientPhone, ClientEmail, ClientCity:* The project will tell us the client and the client will tell us their contact details.
- *ProjectID → SupervisorName → SupervisorPhone :* The project will tell us the supervisor and the supervisor will tell us their phone number.
- *(ProjectID, WorkerName) → SupplierName → materials, equipment and cost:* A worker has only one supplier on a project and everyone with the same supplier on the project has the same materials and equipment.

Which non-key attributes depended on other non-key attributes?

- ClientPhone, ClientEmail, ClientCity depend on ClientName.
- SupervisorPhone depends on SupervisorName.
- MaterialSupplied, MaterialUnitCost, EquipementUsed and EquipmentRentalCost depends on SupplierName (This is on a project)

  We kept the SiteCity and SiteState together because city names can be the same in different states. Workers and Suppliers needed no change.

Which new tables did you create?

New tables	What it stores	Primary key(PK)
3NF_Clients	Client phone, email, city	ClientName
3NF_Supervisors	Supervisor phone	SupervisorName
3NF_Project_Suppliers	The supplier who worked on a particular project	ProjectID, SupplierName
3NF_Project_Supplier_Materials	Materials and unit costs for a supplier and project	ProjectID, SupplierName, MaterialSupplied
3NF_Project_Supplier_Equipment	Equipment and rental costs for a project and supplier	ProjectID, SupplierName, EquipmentUsed

The 3NF_Projects and the 3NF_Project_Assignements were rebuilt without the moved columns. The skills and certifications and supplier phones tables were keyed again and the position columns were dropped. 3NF_Workers and 3NF_Suppliers are unchanged.

How did you use foreign keys to preserve relationships?

- Projects.ClientName points to Clients and Projects.SupervisorName points to Supervisors.

- Project_Assignments points to Projects (ProjectID), WorkerName(Workers) and Project_Suppliers(ProjectID, SupplierName).
- The materials and equipment tables point to Project_Suppliers using (ProjectID, SupplierName).
- The skills and certifications tables point to Project_Assignments using (ProjectID, WorkerName), and Supplier_Phones points to Suppliers.

  When we join the tables together, it will give the same data as 2NF. This shows that no data was lost.

Why is your design now in 3NF?
The tables were already in 2NF, Now in every table each non-key column depends on only the key. Client details are stored under the client, Supervisor details are stored under the supervisor and materials and equipment are stored under the project and supplier. So no non-key column depends on another non-key column.